

Shuai Liu

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Ph.D. Candidate in Embodied AI, Physical Simulation, and Learning-based Control

Researching physics-grounded models, reinforcement learning, multiphysics simulation, and sim-to-real policy learning for embodied intelligence and next-generation robotic systems.

EDUCATION

Shanghai Jiao Tong University, Mechanical Engineering, *Ph.D. Candidate* 2022.9 – 2027.6
Published **11 SCI papers**, including **5 first-author** works on multiphysics computation, nonlinear dynamics, and learning-based control; working on RL control and world-model learning; **National Scholarship Recipient**.
Xi'an Jiaotong University, Mechanical Engineering (Elite Class), *Bachelor of Engineering* 2018.9 – 2022.6
Robotics honor: National First Prize in digital twins and robot-vision integration.

TECHNICAL SKILLS

- **Programming & AI-native Dev:** C/C++ · Python · CUDA · PyTorch · LLM-assisted coding workflows
- **Embodied AI / Robotics:** Reinforcement Learning · Genesis · Newton · Sim-to-Real
- **Physics Simulation:** FEM/FVM · ALE-FSI · Tactile · Multiphysics Coupling · Isaac Sim/Lab
- **GPU / HPC:** CUDA kernel optimization · cuBLAS/cuBLASDx/cuSolver · V100/A100/H100/L40 clusters

INDUSTRY EXPERIENCE

AgiBot, *Embodied Intelligence RL Research Intern* 2026.6 – Present
• Conducting research on **reinforcement-learning frameworks and algorithms** for embodied manipulation, using **RLinf** to support VLA policy post-training, sim-to-real co-training/transfer and simulator interfaces.

Lightwheel, *Embodied Intelligence Simulation Research Intern* 2026.1 – 2026.4
• Built **embodied AI pipelines** across **Genesis**, **Newton**, and **Warp** for contact-rich manipulation data generation, VLA policy evaluation, RL training, and sim-to-real alignment.
• Developed a proprietary tactile-simulator backend in **Newton** and Isaac Lab, reducing FEM mesh size by **5–10×** while preserving contact and sensing fidelity for scalable policy experiments.
• Built a **Genesis** pipeline for Piper arm + Xense tactile sensor with FEM sensing, rigid-soft contact, and IPC coupling; calibrated sim-to-real tactile data and improved throughput by **2.6×**.

Synopsys Inc., *GPU Computing Infra Research Intern* 2025.7 – 2025.10
• Developed and integrated **GPU solver kernels** and linear-algebra primitives for an in-house **simulation infrastructure**, achieving **1.3×** **average speedup** on production workloads across **H100/L40** clusters.
• Developed and optimized batched GEMM kernels based on cuBLASDx, implementing **multi-batch-on-one-block** and **two-stage pipelines** with shared memory tiling. An average speedup of **1.2x** and a maximum of **1.5x** under certain configurations are achieved.

EMBODIED AI INFRASTRUCTURE RESEARCH

Physical Representation Modeling 2022 – Present
• Built high-fidelity multiphysics models for flow-structure, flow-structure-acoustic, and piezoelectric systems, providing physically grounded state spaces for prediction and control.
• Developed **Koopman-theory-based physical-field encoding** and latent state-transition models to predict multimodal multiphysics **world-model** dynamics and support latent-space RL control training.

Learning-based Control And Multimodal Fusion 2022 – Present
• Studied **flow, vibration, non-Hermitian, and RL-based control** for piezoelectric and multiphysics structures, spanning Dreamer-style model-based RL and policy-gradient regimes.
• Developed an **RL control framework** for plate-turbulence interaction, using **Transformer** and **ViT** encoders with **cross-attention** to fuse structural and fluid states.
• Validated **sim-to-real policy transfer** on a circuit-driven **piezoelectric continuum system**; deployed π_0 VLA locally and integrated a physics-consistent solver for contact-rich tasks.

Scalable Physical Simulation For Embodied Data Generation 2022 – Present
• Built a unified framework for **nonlinear continuum mechanics and multiphysics simulation** across CPU/GPU workflows, based on **OpenFOAM + in-house solid solvers + preCICE + RL**, supporting physical data generation, scalable experiments, reproducible control studies, and a **software copyright registration**.

SELECTED PUBLICATIONS

Publication profile: 11 SCI papers, including 5 first-author works; preparing JFM-targeted manuscripts on multimodal RL, physical world-state representation, and sensor-actuator control for embodied AI.

- *Physical–Cyber Multimodal Reinforcement Learning for Flow-Induced Piezoelectric Panel Control in Cylinder Wake*
Manuscript in preparation (target: **J. Fluid Mech.**) · **first-author** — OpenFOAM–preCICE embodied simulation environment with sparse wake probes, distributed piezo sensing, Solid Temporal Encoder, Fluid ViT, cross-attention fusion, and PPO sensor-actuator topology learning.
- *Continuum Multi-Level Piezo-Network Reinforcement Learning for Flow-Induced Vibration Control of Kirchhoff Plates*
Manuscript in preparation (target: **J. Fluid Mech.**) · **first-author** — Continuum physical engine for large-deformation FSI plates, programmable piezoelectric networks, multimodal fluid-structure tokenization, and Dreamer-like MBRL control over distributed actuator gains.
- *Nonlinear flapping and symmetry-breaking bifurcation modulation of a piezoelectric metamaterial beam in viscous flow*
J. Fluid Mech. · 2025 · **first-author** — High-fidelity fluid-structure-electro-control simulation with self-sensing actuator networks, adaptive control, Floquet validation, and Sim-to-Real-ready physical state spaces.
- *Control of flow-structure-acoustic interactions of a non-Hermitian piezoelectric panel-cavity system in supersonic flow*
J. Sound Vib. · 2025 · **first-author** — Controllable multiphysics dynamics, non-Hermitian feedback control, and coupled FSA simulation for physical world representation.
- *Fluid-structure-acoustic responses of a nonlinearly supported rigid circular cylinder in viscous flow*
Phys. Fluids · 2025 · **first-author** — DMD-based multiphysics state-space modeling of nonlinear synchronization, acoustic branches, and physics-consistent world dynamics.
- *Flow-induced vibration and acoustic responses of a rotationally oscillating circular cylinder on a nonlinear elastic mount*
J. Sound Vib. · 2024 · **first-author** — Scalable FEM-FVM physical engine for multimodal flow-structure-acoustic data generation, lock-on analysis, and nonlinear FSI representation learning.
- *Learning Optimal Gain Networks for Vibration Control of Piezoelectric Beams with Distributed Sensor-Actuator Circuits*
ICSV 32 · accepted · **first-author** — Learning-based gain networks for distributed sensor-actuator control in continuum physical systems.
- *Hybrid magnetic coupling and impact enhanced low, adjustable and broadband piezoelectric vibration energy harvesting*
Sensors & Actuators A · 2023 · **first-author** — Smart-structure actuation and energy-coupled physical system design for adaptive devices.
- *Selected co-author works*
J. Sound Vib. 2026 · **Applied Acoustics** 2026 · **Int. J. Multiphase Flow** 2025 · **J. Fluids Struct.** 2026 — Programmable metamaterials, active flexible-surface control, and complex-fluid FSI for embodied simulation.

SELECTED COMPETITION HONORS

- *3rd China University Student Mechanical Engineering Innovation Contest – Intelligent Manufacturing National First Prize · Industrial Vision Track First Prize* — Led robot–vision integration on a smart production line: KUKA manipulator teach-in, trajectory programming, and automated pick-and-place on a physical line. Built a CNN-based part detection and pose-estimation pipeline in OpenCV for vision-guided grasping and in-line quality inspection. Closed the perception–action loop with PLC/OPC UA (C++, open62541) for sensor-triggered motion execution.